Name: _____

WORKSHEET

Derivative of a sum of terms

1 Find the derivatives of the following:

a $x^3 + x^5$ _____

b $x^7 + x$ _____

c $x^2 - x^5$ _____

d $x^6 - x^4$ _____

e $x^{25} + x^{64}$ _____

f $x^{12} - x^3$ _____

g $x^3 + 7$ _____

h $1 - x^9$ _____

2 Find the following

a $\frac{d}{dx}(x^5 + x^8)$ _____

b $\frac{d}{dx}(x^2 - x^4)$ _____

c $\frac{d}{dx}(x^7 + x^3)$ _____

d $\frac{d}{dx}(x^{10} + x^6)$ _____

e $\frac{d}{dx}(x^4 + x)$ _____

f $\frac{d}{dx}(-x^3 + x^5)$ _____

g $\frac{d}{dx}(-x^3 + 14)$ _____

h $\frac{d}{dx}(-x^8 - x)$ _____

© Cengage Learning Australia Pty Ltd 2013

3 Find the derivative of the following

a $x^6 + x^3$ _____

b $s^7 + s^2$ _____

c $t^8 + t^4$ _____

d $y^9 - y$ _____

e $w^6 - w^{10}$ _____

f $p^3 - 1$ _____

g $-k^5 - k^4$ _____

h $m^{11} - m^{19}$ _____

4 Find the following

a $\frac{d}{dx}(x^3 - x^2)$ _____

b $\frac{d}{dz}(z^7 + z^3)$ _____

c $\frac{d}{dq}(q^5 - q)$ _____

d $\frac{d}{dr}(-r^7 + r^8)$ _____

e $\frac{d}{d\$}(\$^6 + \$^{11})$ _____

f $\frac{d}{dx}(x^3 + 4)$ _____

g $\frac{d}{dx}(x^5 + c)$ _____

h $\frac{d}{dx}(x^2 - y)$ _____

© Cengage Learning Australia Pty Ltd 2013

Answers

1 **a** $3x^2 + 5x^4$

b $7x^6 + 1$

c $2x + 5x^4$

d $6x^5 - 4x^3$

e $25x^{24} + 64x^{63}$

f $12x^{11} - 3x^2$

g $3x^2$

h $-9x^8$

2 **a** $5x^4 + 8x^7$

b $2x - 4x^3$

c $7x^6 + 3x^2$

d $10x^9 + 9x^5$

e $4x^3 + 1$

f $-3x^2 - 5x^4$

g -1

h $-8x^7 - 1$

3 **a** $6x^5 + 3x^2$

b $7s^6 + 2s$

c $8t^7 + 4t^3$

d $9y^8 - 1$

e $6w^5 - 10w^9$

f $3p^2$

g $-5k^4 - 4k^3$

h $11m^{10} - 19m^{18}$

4 **a** $3x^2 - 2x$

b $7z^6 + 3z^2$

c $5q^4 - 1$

d $-7r^6 + 8r^7$

e $6\$^5 + 11\$^{10}$

f $3x^2$

g $5x^4$

h $2x$

© Cengage Learning Australia Pty Ltd 2013